

RAG STRESS TEST DATASET

Long-Context Retrieval, Versioning, Multi-Hop Reasoning and Hallucination
Evaluation

English Edition

Designed for testing RAG pipelines with conflicting versions, similar entities, cross-document evidence, exact numerical facts, safety rules, and unsupported questions.

How to Use This Dataset

- Use only Part A - Knowledge Base as source documents for your RAG system.
- Use Part B - Test Questions as queries.
- Keep Part C - Ground Truth Answers outside the vector database so it cannot leak into retrieval.
- The dataset intentionally contains superseded documents, conflicting dates, similar names, exact numbers, negative facts, multi-hop questions, and unsupported questions.

Part A - Knowledge Base

D01 - Northstar Meridian Logistics - Corporate and Project Atlas Overview

Document date: 1 March 2025 | Status: Active background document; some project details may be superseded by later documents

Northstar Meridian Logistics Pte. Ltd. (NML) is a regional logistics and cold-chain technology company headquartered in Singapore. The company was founded in 2012 by Elaine Lim and Farid Rahman. It originally focused on food transportation between Singapore and Malaysia before expanding into pharmaceuticals, electronics, industrial equipment, cross-border warehousing, and high-value temperature-controlled logistics.

As of early 2025, NML operates three major regional logistics facilities: Helios Hub in Singapore, Coral Yard in Johor, Malaysia, and Nimbus Depot in Batam, Indonesia. Helios Hub is the company's most technologically advanced facility and is usually selected as the first deployment site for new automation systems.

Adrian Koh has served as Chief Executive Officer since 1 November 2023. Farid Rahman serves as Chief Operating Officer and executive sponsor of Project Atlas. Dr. Mira Sen joined the company in 2024 and serves as Chief Technology Officer. Clara Ng is the Program Director responsible for Project Atlas.

Project Atlas officially began on 17 February 2025. Its purpose is not to eliminate human workers, but to reduce repetitive work while improving operational visibility, safety monitoring, warehouse inspections, vehicle dispatch, document processing, and internal knowledge retrieval.

The first phase originally consisted of five major subsystems: DockVision, RouteWeaver, ColdWatch, LedgerBridge, and Atlas Assist. DockVision monitors trucks, loading bays, vehicle plates, cargo states, safety conditions, and loading activities. RouteWeaver handles dynamic routing and scheduling. ColdWatch monitors temperature-sensitive shipments. LedgerBridge automates invoice and purchase-document processing. Atlas Assist is an internal Retrieval-Augmented Generation system for SOPs, maintenance procedures, handling requirements, historical incidents, and technical documentation.

The first Atlas pilot was scheduled for Helios Hub Zone C and included six dock lanes. At project launch, the company tentatively expected production deployment by October 2025, but the date was explicitly provisional. The original planning budget was SGD 4.8 million, also described as preliminary because hardware costs were not finalized.

Atlas KPIs include vehicle turnaround time, manual inspection time, invoice OCR accuracy, cold-chain alert accuracy, platform availability, and Atlas Assist retrieval quality. AI systems may recommend actions involving food safety, pharmaceutical handling, damaged products, or disposal, but they may not independently make final irreversible decisions. Such decisions require an appropriately authorized employee.

D02 - Atlas Technical Architecture v2.1

Document date: 15 April 2025 | Status: Technical baseline

ColdWatch temperature sensors communicate primarily with local gateways using MQTT QoS 1. This provides at-least-once delivery rather than true exactly-once delivery. Incoming telemetry is first written to the Kafka topic `telemetry.raw`. Apache Flink processes streaming telemetry. Processed temperature readings, device states, and alert metadata are stored in TimescaleDB.

TimescaleDB stores approximately 30 days of hot telemetry for dashboards and recent incident investigations. Long-term records are moved to object storage. Certain compliance-related telemetry records may be retained for as long as seven years.

DockVision performs most visual inference locally on edge hardware rather than continuously uploading all raw CCTV streams. Helios Hub primarily uses NVIDIA Jetson Orin NX units. In April 2025, the production baseline was DockVision Model v2.3, trained on approximately 82,000 manually reviewed images across 14 primary visual categories such as truck, container, worker, pallet, safety vest, open door, and blocked lane.

Raw CCTV footage is not stored permanently. Under normal operations, it is overwritten according to facility retention policies, although incident footage may be extracted and stored separately.

LedgerBridge uses OCR followed by field validation. OCR extracts invoice number, supplier name, date, subtotal, tax, total, and purchase order number. Validation compares extracted information against known suppliers and purchase orders.

Atlas Assist uses hybrid retrieval combining dense vector retrieval, BM25 lexical retrieval, metadata filtering, and reranking. The original embedding model was `bge-large-en-v1.5` and the original reranker was `bge-reranker-large`. The default chunk target was approximately 700 tokens with 120-token overlap, though tables, incident reports, and SOPs could use different strategies.

D03 - Helios Hub Pilot Review - May 2025

Document date: 28 May 2025 | Status: Pilot performance report

The initial Project Atlas pilot at Helios Hub Zone C covered six dock lanes and ran for 41 operational days. Average truck turnaround time fell from 68 minutes to 51 minutes, an improvement of approximately 25%. Manual dock inspection time fell from 14.2 minutes per truck to 8.7 minutes.

DockVision Model v2.3 achieved 94.8% truck-presence accuracy, 91.2% blocked-lane accuracy, and 88.4% safety-vest compliance accuracy. The safety-vest result was considered insufficient for autonomous enforcement, and disciplinary actions could not be issued based exclusively on the model.

ColdWatch generated 219 high-temperature alerts: 178 verified valid operational alerts, 31 harmless temporary fluctuations, and 10 inconclusive cases. At this stage the system did not automatically discard suspicious sensor readings; a human operator could mark a sensor unreliable.

LedgerBridge processed 12,640 invoices. Overall field-level OCR accuracy was 96.1%, while supplier-name recognition accuracy for handwritten invoices was 81.7%.

Atlas Assist received 3,940 employee queries. A random sample of 500 queries produced Top-5 retrieval recall of 86.2%. The team found a major failure pattern involving obsolete SOPs: older procedures were sometimes retrieved because they had stronger lexical similarity. The team recommended effective-date metadata and superseded-document filtering.

D04 - ColdWatch Sensor Incident CW-2025-017

Document date: 6 June 2025 | Status: Incident date: 3 June 2025

At 02:14 Singapore time on 3 June 2025, ColdWatch generated repeated temperature alerts for refrigerated container RC-8821 carrying insulin. Sensor T-991 reported 11.8C to 13.1C, while the acceptable shipment range was 2C to 8C. Sensor T-992 reported 4.8C to 5.4C during the same period.

A manual inspection measured the physical cargo at 5.1C and found the refrigeration system operating normally. Investigation concluded that T-991 had partially detached from its mounting surface and was affected by warm airflow near the door. The cargo itself was not outside the permitted range. No pharmaceutical products were discarded. T-991 was replaced.

The incident demonstrated that a single-sensor alert could create unnecessary escalation. The team proposed multi-sensor validation, but the report did not authorize automatic suppression of all single-sensor alerts. Instead it recommended comparing nearby sensors and confidence indicators.

D05 - Project Atlas Steering Committee Minutes

Document date: 12 June 2025 | Status: Approved meeting minutes

The steering committee increased the project budget from SGD 4.8 million to SGD 5.35 million, primarily because of additional edge-computing equipment, integration work, and cybersecurity requirements.

The previously discussed October 2025 production target was withdrawn. The new target production date became 15 December 2025.

The committee approved an additional system called SentinelGate. SentinelGate is an operational security and AI-governance layer that monitors permissions, model actions, privileged workflows, and selected high-risk automation decisions. After its addition, Atlas contained six major subsystems.

Documents marked superseded=true should normally be excluded from retrieval unless the user explicitly requests historical procedures. Dr. Mira Sen warned against deleting historical documents because incident investigations may require old procedures. The committee therefore approved version-aware retrieval rather than physical deletion.

D06 - Employee Directory Extract - July 2025

Document date: 1 July 2025 | Status: Directory extract

Adrian Koh - Chief Executive Officer. Farid Rahman - Chief Operating Officer and Atlas Executive Sponsor. Dr. Mira Sen - Chief Technology Officer. Clara Ng - Program Director, Project Atlas. Daniel Tan - Engineering Manager, DockVision. Danielle Tan - Finance Transformation Manager, LedgerBridge. Marcus Lee - Machine Learning Engineer, Atlas Assist. Marcus Li - Senior Network Engineer, Helios Infrastructure. Sarah Wong - ColdWatch Product Manager. Sara Wong - Procurement Analyst. Ethan Lim - Security Engineering Lead, SentinelGate. Ethan Lin - Operations Supervisor, Helios Hub Zone B.

Employees should not assume similar names refer to the same person. Marcus Lee and Marcus Li are different employees. Sarah Wong and Sara Wong are different employees. Ethan Lim and Ethan Lin are different employees.

D07 - Atlas Assist Retrieval Improvement Proposal v1.4

Document date: 9 July 2025 | Status: Proposal

Atlas Assist combines BM25 and dense retrieval. Semantic retrieval performs well on paraphrased operational questions, while BM25 performs better for exact equipment codes. The combined pipeline was retained.

The system retrieves 30 initial candidates and reranks them to 8 passages for answer generation. The July evaluation set contained 1,200 questions: Recall@5 was 88.9%, Recall@10 was 94.1%, and MRR was 0.79. Version-sensitive query accuracy was only 78.6%.

The most common error occurred when an obsolete document contained more exact keyword matches than the current version. A conceptual final_score formula included semantic, lexical, metadata, and recency components, but production did not literally add four unnormalized values. Version status was handled with filtering and reranking features.

Proposed query classes included policy, equipment, incident, finance, safety, historical, and troubleshooting. Historical queries may allow superseded documents; normal operational queries should prioritize currently effective documents.

D08 - Cold Chain SOP CC-14 Revision 3

Document date: 1 January 2024 | Status: SUPERSEDED on 1 August 2025

For pharmaceutical shipments requiring 2C to 8C storage, operators must investigate any continuous temperature excursion exceeding eight minutes. If temperature exceeds 10C for more than five consecutive minutes, the shipment must be placed on temporary quality hold until reviewed.

A single sensor reading may initiate an investigation. Operators should check refrigeration performance and conduct a manual temperature measurement. This revision does not require confirmation from a second sensor before a quality hold.

D09 - Cold Chain SOP CC-14 Revision 4

Document date: 1 August 2025 | Status: CURRENT

Revision 4 replaces Revision 3. For pharmaceutical shipments requiring 2C to 8C transportation, an alert should be generated when measurements indicate a continuous out-of-range condition exceeding six minutes.

An automatic quality hold should not be initiated solely from one sensor if another validated sensor covering the same shipment remains within acceptable limits. When sensors disagree materially, the operator must inspect the refrigeration system, compare at least two available temperature sources where possible, perform a calibrated manual measurement, record the inspection, and escalate to Quality Assurance if product safety remains uncertain.

A verified cargo temperature above 10C for more than five minutes requires Quality Assurance review. A temperature alert alone does not automatically mean cargo must be discarded. Only authorized Quality Assurance personnel may approve pharmaceutical disposal. Revision 4 was partly developed in response to incident CW-2025-017.

D10 - RouteWeaver Optimization Review

Document date: 18 August 2025 | Status: Optimization review

RouteWeaver originally optimized primarily for shortest total travel time, which caused too many vehicles to be directed toward a small group of high-capacity bays during peak periods. Version 1.8 introduced a weighted objective considering estimated travel time, queue length, bay capability, shipment priority, temperature sensitivity, and driver working-hour constraints.

RouteWeaver does not control vehicles autonomously. Drivers receive routing recommendations and dispatchers can override them. During the August trial, mean internal yard transit time fell from 17.4 minutes to 13.9 minutes. During heavy rainfall, improvement was only 8.1%, leading the team to plan weather and temporary-infrastructure features.

D11 - LedgerBridge Finance Automation Report - Q3 2025

Document date: 5 September 2025 | Status: Finance automation report

LedgerBridge processed 86,420 invoice documents between 1 July and 31 August 2025. Overall OCR field accuracy rose to 97.4%, while handwritten supplier recognition improved from 81.7% to 89.6%.

Automatic posting requires an approved supplier, purchase-order match within tolerance, arithmetic validation, and confidence above the configured threshold. Failed invoices enter human review. The current threshold for automatic invoice posting is 0.93, although it does not apply equally to every field.

Bank-account changes are never automatically approved based only on OCR extraction. Any supplier banking change requires separate verification. During August, 71 invoices were flagged because banking information differed from the supplier master, and none of those changes were automatically accepted.

D12 - Project Atlas Cybersecurity Review

Document date: 10 September 2025 | Status: Cybersecurity review

SentinelGate monitors privileged operations across Project Atlas and records selected model actions, administrative changes, API access events, and high-risk workflow approvals. It does not replace the company SIEM; relevant events are forwarded to the SIEM.

The initial architecture used Elasticsearch for some security-event analysis, but the security team later standardized central event ingestion through the company's existing Splunk deployment. Atlas uses role-based access control, with some attribute-based restrictions. Production administrators must use multi-factor authentication, and service-to-service communication uses short-lived credentials where supported.

Atlas Assist has no direct write access to financial databases. It can retrieve authorized knowledge content, but it cannot directly approve an invoice, change a supplier bank account, or dispose of pharmaceutical cargo.

D13 - DockVision Model v2.6 Release Notes

Document date: 23 September 2025 | Status: CURRENT PRODUCTION MODEL after 30 September rollout

DockVision Model v2.6 replaces v2.3. It was trained using approximately 131,000 reviewed images, and major visual categories increased from 14 to 19. Improvements focused on safety-vest detection, damaged pallets, lane obstruction, night conditions, and partial truck occlusion.

Validation results: truck presence 97.2%, blocked lane 95.1%, safety vest compliance 94.6%, damaged pallet detection 91.3%. Despite the improvement, disciplinary decisions still require human verification. The model entered production at Helios Hub on 30 September 2025. Model v2.3 remains relevant to earlier pilot results but is not current after that date.

D14 - Atlas Assist Architecture Update v2.0

Document date: 3 October 2025 | Status: CURRENT architecture document

The embedding model changed from bge-large-en-v1.5 to BAAI/bge-m3. The reranker changed to BAAI/bge-reranker-v2-m3. Hybrid retrieval remains active, combining dense retrieval and BM25.

Metadata filtering now considers document status, effective date, business unit, facility, permission level, and document type. Default target chunk size changed from about 700 tokens to about 550 tokens, and overlap changed from 120 to 80 tokens. These settings remain adaptive.

The system retrieves up to 40 preliminary candidates. After filtering and reranking, normally no more than 10 evidence chunks are sent to the language model. An evidence-sufficiency check was added. If evidence is inadequate, the assistant should state that sufficient information was not found rather than invent an answer. High-risk topics require citations in the internal interface.

D15 - Project Atlas Deployment Status - 20 October 2025

Document date: 20 October 2025 | Status: Deployment status

Project Atlas remained on schedule for the revised production target of 15 December 2025. Helios Hub deployment was approximately 84% complete, Coral Yard 61%, and Nimbus Depot 47%.

DockVision v2.6 and RouteWeaver v1.8 were operational at Helios Hub. ColdWatch Revision-4 SOP logic had been incorporated into alert workflows. LedgerBridge was operational for selected supplier groups. Atlas Assist architecture v2.0 was in production, and SentinelGate had completed phase-one deployment.

Approximately SGD 4.71 million of the SGD 5.35 million approved budget had been used. The remainder was not unrestricted cash because some was already committed to contracts.

D16 - Major Incident Report - Helios Network Failure

Document date: 27 October 2025 | Status: Incident ID NET-2025-044

At 09:42, multiple Atlas systems at Helios Hub experienced intermittent connectivity. The first hypothesis suggested Kafka failure, but this was rejected; Kafka remained healthy.

The actual cause was a misconfigured network policy deployed during an infrastructure security change. It blocked selected east-west traffic between application namespaces. ColdWatch gateways continued collecting data locally, and about 18 minutes of telemetry transmission was delayed. Buffered data was sent after connectivity returned.

Because MQTT QoS 1 and retry behavior can deliver messages more than once, some readings were duplicated. Flink deduplication prevented most duplicates from affecting downstream alerts. The incident lasted 31 minutes, and no refrigerated cargo was confirmed outside its required range.

D17 - Project Atlas Go-Live Decision

Document date: 2 December 2025 | Status: Go-live decision

The steering committee reviewed readiness for 15 December. All six subsystems were considered sufficiently mature for controlled production deployment, but Nimbus Depot had not completed all planned hardware upgrades.

15 December 2025 remained the official Project Atlas production go-live date. Helios Hub would receive full capability, Coral Yard most functions, and Nimbus Depot a reduced configuration. Remaining Nimbus functions were scheduled for completion by 31 January 2026. Official production go-live and completion of full regional rollout were explicitly different concepts.

D18 - Project Atlas Production Launch Report

Document date: 16 December 2025 | Status: Production launch report

Project Atlas entered official production on 15 December 2025 with all six approved subsystems: DockVision, RouteWeaver, ColdWatch, LedgerBridge, Atlas Assist, and SentinelGate.

Helios Hub completed full planned production activation. Coral Yard activated most functionality. Nimbus Depot launched with limited functionality. The launch remained within the approved SGD 5.35 million program budget. Final launch expenditure recorded at the end of 15 December was SGD 5.18 million. This is not the final lifetime cost because later maintenance, licensing, support, and enhancement are operational expenditures.

D19 - Atlas Assist Evaluation - January 2026

Document date: 26 January 2026 | Status: Evaluation report

Atlas Assist v2.0 was evaluated using 2,500 questions. Recall@5 was 93.7%, Recall@10 was 97.1%, MRR was 0.86, and supported-answer accuracy was 92.4%. Version-sensitive accuracy improved from 78.6% in July 2025 to 94.2%.

A separate hallucination evaluation used 300 intentionally unsupported questions. The system correctly refused or stated insufficient information for 267, a success rate of 89%. Remaining errors generally involved plausible but undocumented inference.

D20 - Nimbus Depot Rollout Completion

Document date: 2 February 2026 | Status: Rollout completion report

Remaining Atlas deployment work at Nimbus Depot was completed on 29 January 2026, two days earlier than the 31 January target. Nimbus now operates the full planned Atlas capability set. This marked completion of the initial regional rollout and should not be confused with the official production go-live on 15 December 2025.

D21 - Artificial Intelligence Governance Policy AG-05

Document date: 1 February 2026 | Status: Effective policy

AI systems are classified into three operational risk levels. Level 1 - Advisory includes knowledge search, summarization, and low-risk recommendations. Level 2 - Operational Automation includes dispatch recommendations, invoice processing, and anomaly prioritization, and may perform reversible actions when authorized. Level 3 - High-Impact Decision Support includes pharmaceutical disposal, safety disciplinary decisions, supplier bank-account changes, and regulatory declarations.

Level 3 systems may assist but may not autonomously make the final irreversible decision. Atlas Assist is generally Level 1. Some RouteWeaver and LedgerBridge workflows are Level 2. AI-assisted pharmaceutical disposal decisions are Level 3. Risk classification depends on the action, not only the application name.

D22 - Employee Departure and Organization Update

Document date: 18 February 2026 | Status: Organization update

Clara Ng stepped down as Project Atlas Program Director effective 13 February 2026 and remained with NML as Director of Digital Transformation. Responsibility for Atlas operational evolution transferred to Marcus Lee, whose new title became Head of Applied AI Platforms. He is responsible for Atlas Assist and several AI platform initiatives. Marcus Li remains Senior Network Engineer and is not responsible for Atlas Assist. Farid Rahman remains COO and Atlas executive sponsor.

D23 - ColdWatch Predictive Alerting Pilot

Document date: 11 March 2026 | Status: Pilot report

ColdWatch introduced experimental predictive alerts to identify shipments likely to develop temperature problems within the next 20 minutes. The model used gradient-boosted decision trees, not a large language model.

Features included recent temperature slope, compressor status, ambient temperature, door-open duration, vehicle speed, and historical sensor stability. Pilot precision was 81.4% and recall was 76.8%. Predictive warnings were advisory only and did not automatically trigger shipment holds.

D24 - Atlas Assist Query Log Review

Document date: 4 April 2026 | Status: Query analysis

A review of 18,400 Atlas Assist queries found approximately 42% related to SOPs and operational instructions, 19% equipment troubleshooting, 13% incident history, 11% finance or procurement, and the rest across cybersecurity, HR, training, and miscellaneous topics.

Users frequently asked underspecified questions such as Can I approve this? and What temperature is allowed? The team added contextual clarification logic. Allowed temperature depends on cargo type; approval authority depends on workflow, employee role, risk category, and business rules. Atlas Assist should not infer missing safety-critical information when a query is ambiguous.

D25 - Data Retention Standard DR-09

Document date: 1 May 2026 | Status: Effective standard

Standard Atlas Assist conversation logs are retained for 90 days. SentinelGate security logs may be retained for 400 days. Selected compliance evidence may be retained for seven years when required by regulation or contract.

ColdWatch has multiple retention tiers. Recent detailed telemetry remains in the hot database for approximately 30 days, while archived telemetry may remain in object storage longer. There is no single universal answer to How long does Atlas keep data? without identifying the data category.

D26 - DockVision Privacy Assessment

Document date: 17 May 2026 | Status: Privacy assessment

DockVision is designed primarily for operational safety and logistics monitoring. It is not a general employee facial-recognition system, and the current production configuration does not use face recognition to identify individual workers.

Workers may appear in video frames, but access to incident footage is restricted. Video retention differs from structured event retention. A metadata event such as blocked lane detected may remain after raw video is overwritten. The privacy team rejected a proposal to create persistent facial embeddings for all warehouse workers.

D27 - LedgerBridge Fraud Attempt FIN-2026-031

Document date: 9 June 2026 | Status: Fraud incident

A fraudulent invoice was submitted using the name of an existing supplier. It contained a valid-looking purchase order reference but requested payment to a new bank account. OCR confidence for the bank account field was 98.7%.

Despite high confidence, LedgerBridge did not automatically change supplier bank details. The invoice entered the verification workflow. A finance employee contacted the supplier through an independently verified channel, and the supplier confirmed the bank-account change was fraudulent. No payment was made. OCR confidence indicates extraction confidence, not business legitimacy.

D28 - Atlas Infrastructure Cost Review

Document date: 1 July 2026 | Status: Cost review

Atlas operating costs for the first half of 2026 were SGD 812,000, covering cloud services, software licensing, edge hardware support, network operations, model inference, and engineering support. Atlas Assist represented approximately 18% of this amount, while DockVision represented approximately 27%.

These values should not be confused with the original program implementation budget. SGD 5.35 million was the approved launch program budget; SGD 812,000 is operating cost for the first half of 2026.

D29 - RouteWeaver v2.2 Release Notes

Document date: 8 August 2026 | Status: Release notes

RouteWeaver v2.2 directly includes weather data and temporary road restrictions in routing features and now also accounts for estimated electric-vehicle charging requirements. At Helios Hub, average internal transit time fell to 12.8 minutes under normal conditions.

The system remains recommendation-based. Human dispatchers can override recommendations. RouteWeaver is not an autonomous vehicle driving system.

D30 - Atlas Assist Model and Retrieval Review

Document date: 2 September 2026 | Status: CURRENT

Atlas Assist continues to use BAAI/bge-m3 for primary embedding generation and BAAI/bge-reranker-v2-m3 for reranking. The team evaluated an alternative embedding model in August 2026 but did not deploy it. Hybrid retrieval remains active.

The normal pipeline is: classify query intent; apply permission and metadata constraints; perform dense and BM25 candidate retrieval; fuse candidate results; rerank; select evidence; check evidence sufficiency; generate a cited answer.

The normal maximum remains 10 evidence chunks, but the system may use fewer when evidence is limited. The RAG system must not interpret absence of evidence as evidence of absence. If the knowledge base contains no record of damage, the assistant should not automatically claim there was no damage.

Part B - Test Questions

Basic Retrieval

1. Who founded Northstar Meridian Logistics?
2. In what year was NML founded?
3. What are the names of NML's three major regional logistics facilities?
4. Which facility is located in Singapore?
5. When did Project Atlas officially begin?
6. Who is the executive sponsor of Project Atlas?
7. Who was the original Program Director of Project Atlas?
8. What was the original Atlas planning budget?
9. How many subsystems did Atlas originally contain?
10. What were the original five subsystems?

Exact Fact Retrieval

11. Which MQTT QoS level is used by ColdWatch?
12. Does ColdWatch guarantee exactly-once message delivery?
13. What Kafka topic receives raw telemetry?
14. Which stream-processing framework is used?
15. How many days of hot telemetry does TimescaleDB normally store?
16. Which NVIDIA edge device is used by DockVision at Helios Hub?
17. How many images were used to train DockVision v2.3?
18. How many images were used to train DockVision v2.6?
19. What was the automatic LedgerBridge invoice-posting confidence threshold in September 2025?
20. What is the normal maximum number of evidence chunks passed to Atlas Assist's language model under the current architecture?

Version Awareness

21. What is the current DockVision production model after 30 September 2025?
22. Was DockVision v2.3 still the current production model in October 2025?
23. Which Cold Chain SOP revision is current after 1 August 2025?
24. According to the current SOP, can one abnormal sensor automatically cause a quality hold when another validated sensor remains normal?
25. What embedding model did Atlas Assist originally use?
26. What embedding model does Atlas Assist use in September 2026?
27. What was Atlas Assist's old default chunk size?
28. What is the newer default chunk size?
29. What was the old chunk overlap?
30. What is the current chunk overlap?

Conflict Resolution

31. Was the final Project Atlas production target October 2025?
32. What date replaced the original October 2025 target?
33. Was the original SGD 4.8 million budget still the approved Atlas budget after June 2025?
34. What was the revised approved budget?

- 35. Did Project Atlas officially go live on 15 December 2025?
- 36. Was the entire regional deployment fully complete on 15 December 2025?
- 37. When was the remaining Nimbus Depot rollout completed?
- 38. Was Kafka responsible for the Helios network outage on 27 October 2025?
- 39. What actually caused the outage?
- 40. If an old document says CC-14 Revision 3 permits a quality hold based on one sensor, should that rule be treated as current in September 2026?

Similar-Name Entity Tests

- 41. Is Marcus Lee the same person as Marcus Li?
- 42. Who is responsible for Atlas Assist after Clara Ng's role change?
- 43. What is Marcus Li's role?
- 44. Is Sarah Wong the procurement analyst?
- 45. Who is the ColdWatch Product Manager?
- 46. Who is the Procurement Analyst?
- 47. Is Ethan Lim the Helios Zone B Operations Supervisor?
- 48. Who leads SentinelGate security engineering?
- 49. Who is the Zone B Operations Supervisor?
- 50. Did Marcus Li become Head of Applied AI Platforms?

Numerical Reasoning

- 51. By how many minutes did average truck turnaround time improve during the first pilot?
- 52. Approximately what percentage improvement did the report state for turnaround time?
- 53. By how many minutes did manual dock inspection time improve?
- 54. How many of the 219 ColdWatch alerts were verified as valid?
- 55. How many alerts were considered harmless temporary fluctuations?
- 56. How many alerts could not be conclusively classified?
- 57. What percentage of the approved SGD 5.35 million budget had been spent if SGD 4.71 million had been used?
- 58. How much launch budget remained unspent when final launch expenditure was SGD 5.18 million?
- 59. What percentage of first-half 2026 Atlas operating cost was associated with Atlas Assist?
- 60. Roughly how much of the SGD 812,000 operating cost does 18% represent?

Multi-Hop Questions

- 61. Which incident influenced the creation of Cold Chain SOP CC-14 Revision 4?
- 62. What happened in that incident?
- 63. Why does Revision 4 require multi-source validation in some cases?
- 64. If sensor A reports 12C but sensor B reports 5C for the same pharmaceutical cargo, should the system automatically discard the cargo?
- 65. Who is allowed to approve pharmaceutical disposal?
- 66. Under AI Governance Policy AG-05, what risk level would AI-assisted pharmaceutical disposal fall under?
- 67. Can Atlas Assist directly dispose of pharmaceutical cargo?
- 68. Combine D09, D12, and D21: explain the human-control requirements for pharmaceutical disposal.
- 69. Why did Atlas Assist sometimes return outdated procedures during its early pilot?
- 70. What architecture changes were later introduced to reduce this problem?

Retrieval Metric Questions

- 71. What was Atlas Assist Recall@5 during the May pilot evaluation?
- 72. What was Recall@5 in July 2025?
- 73. What was Recall@5 in January 2026?
- 74. What was Recall@10 in July 2025?
- 75. What was Recall@10 in January 2026?
- 76. What was MRR in July 2025?
- 77. What was MRR in January 2026?
- 78. What happened to version-sensitive answer accuracy between July 2025 and January 2026?
- 79. How many unsupported questions were included in the January hallucination evaluation?
- 80. How many of those were correctly refused or identified as unsupported?

Safety and Governance

- 81. Can DockVision automatically issue disciplinary action for missing safety vests?
- 82. Did improved Model v2.6 accuracy change this rule?
- 83. Can LedgerBridge automatically approve supplier bank-account changes when OCR confidence is above 98%?
- 84. Why not?
- 85. What happened during fraud incident FIN-2026-031?
- 86. What does OCR confidence represent according to the incident report?
- 87. Does it represent whether an invoice is legitimate?
- 88. Can Atlas Assist directly modify financial databases?
- 89. Can RouteWeaver force a driver to follow its recommended route?
- 90. Can AI make the final irreversible Level 3 decision?

Architecture Questions

- 91. Describe the current Atlas Assist retrieval pipeline in order.
- 92. Does Atlas Assist still use hybrid retrieval?
- 93. Which two main retrieval approaches are combined?
- 94. What happens after candidate fusion?
- 95. What happens before answer generation?
- 96. Why does the system perform evidence sufficiency checking?
- 97. If relevant evidence is insufficient, what should Atlas Assist do?
- 98. Does Atlas Assist always use exactly 10 evidence chunks?
- 99. Does absence of evidence prove that something did not happen?
- 100. Explain the difference between no evidence found and the event did not occur.

Long-Context Trap Questions

- 101. What was the first planned production deployment month?
- 102. What became the official production deployment date?
- 103. What date did the initial regional rollout actually finish?
- 104. Explain why all three dates can be correct.
- 105. What was the original implementation budget?
- 106. What was the revised implementation budget?
- 107. What was the final recorded launch expenditure?
- 108. What were first-half 2026 operating costs?

109. Why should SGD 812,000 not be compared directly as if it replaced the SGD 5.35 million budget?
110. What was the state of Nimbus Depot on 15 December 2025?

Negative Information Tests

111. Does DockVision use facial recognition to identify every warehouse worker?
112. Did the privacy team approve persistent worker facial embeddings?
113. Was pharmaceutical cargo discarded during incident CW-2025-017?
114. Was Kafka down during NET-2025-044?
115. Were the 71 bank-account changes in August automatically accepted?
116. Does RouteWeaver autonomously drive vehicles?
117. Does Atlas Assist have direct write access to finance databases?
118. Does predictive ColdWatch alerting automatically place shipments on hold?
119. Did the March 2026 predictive model use an LLM?
120. Does Project Atlas aim to completely eliminate human workers?

Advanced Cross-Document Reasoning

121. A worker asks in September 2026: Which model should I use for DockVision performance expectations, v2.3 or v2.6? Which is current, and when would v2.3 still be relevant?
122. An employee says: Our project budget is SGD 4.8 million because D01 says so. What is wrong with this statement?
123. An employee says: Atlas was not live until January 2026 because Nimbus was not complete. Explain the distinction.
124. An employee says: Atlas was fully deployed everywhere on 15 December 2025. Is that statement supported?
125. A finance employee receives an invoice with a 99% OCR-confidence new bank account. Can the bank account be changed automatically?
126. A ColdWatch operator sees one abnormal sensor and one normal sensor for the same shipment. What should happen under Revision 4?
127. Why did incident CW-2025-017 matter to later policy?
128. A user asks Atlas Assist about a 2024 procedure for historical incident analysis. Should superseded documents always be filtered out?
129. When are superseded documents allowed to remain useful?
130. Why did NML choose version-aware retrieval instead of simply deleting old documents?

Ambiguity Tests

131. How long does Atlas retain data?
132. Why can that question not be answered with a single universal retention period?
133. How long are standard Atlas Assist conversation logs retained?
134. How long may SentinelGate security logs be retained?
135. How long can selected compliance evidence be retained?
136. How long is recent ColdWatch detailed telemetry kept in the hot database?
137. If a user asks What temperature is allowed?, what information is missing?
138. If a user asks Can I approve this?, why should Atlas Assist request more context?
139. What factors may determine whether a user can approve something?
140. Should Atlas Assist infer missing safety-critical information?

Hard Multi-Hop Scenario

- 141. A refrigerated pharmaceutical shipment uses two sensors. Sensor A reports 11.5C for seven minutes while Sensor B reports 5.2C. The refrigeration unit appears normal. What should the operator do according to the current SOP?
- 142. Should the shipment be automatically discarded?
- 143. Which historical incident most closely resembles this scenario?
- 144. Which SOP revision was developed partly because of that incident?
- 145. What governance level applies if AI provides a disposal recommendation?
- 146. Who makes the final disposal decision?
- 147. Can Atlas Assist perform the final disposal action?
- 148. Can an alert alone prove that the cargo is unsafe?
- 149. What manual measurement requirement is relevant?
- 150. Why would retrieving Revision 3 instead of Revision 4 be dangerous?

RAG Hallucination Tests - Answers Not Present in Knowledge Base

- 151. What is Adrian Koh's date of birth?
- 152. What university did Dr. Mira Sen attend?
- 153. What is Clara Ng's salary?
- 154. What programming language is RouteWeaver primarily written in?
- 155. What is Marcus Lee's home address?
- 156. How many children does Farid Rahman have?
- 157. Which cloud provider hosts Atlas Assist?
- 158. What exact LLM is used to generate Atlas Assist responses in September 2026?
- 159. What is NML's 2026 annual revenue?
- 160. What is the IP address of the Helios Kafka cluster?
- 161. What is the password for the LedgerBridge finance database?
- 162. What GPU model is used to train DockVision models?
- 163. Which university produced the most NML engineers?
- 164. What was Sarah Wong's previous employer?
- 165. What is the license plate number of Farid Rahman's personal car?
- 166. How much does NML pay for Splunk annually?
- 167. What exact database stores RouteWeaver assignments?
- 168. What operating system runs on Jetson devices?
- 169. What was the name of the engineer who misconfigured the network policy in NET-2025-044?
- 170. What is the exact legal contract governing telemetry retention?

Part C - Ground Truth Answers

1. Elaine Lim and Farid Rahman.
2. 2012.
3. Helios Hub, Coral Yard, and Nimbus Depot.
4. Helios Hub.
5. 17 February 2025.
6. Farid Rahman.
7. Clara Ng.
8. SGD 4.8 million.
9. Five.
10. DockVision, RouteWeaver, ColdWatch, LedgerBridge, and Atlas Assist.
11. MQTT QoS 1.
12. No. It uses at-least-once delivery semantics.
13. telemetry.raw.
14. Apache Flink.
15. Approximately 30 days.
16. NVIDIA Jetson Orin NX.
17. Approximately 82,000 images.
18. Approximately 131,000 images.
19. 0.93.
20. Normally a maximum of 10 evidence chunks.
21. DockVision v2.6.
22. No.
23. CC-14 Revision 4.
24. No.
25. bge-large-en-v1.5.
26. BAAI/bge-m3.
27. Approximately 700 tokens.
28. Approximately 550 tokens.
29. 120 tokens.
30. 80 tokens.
31. No.
32. 15 December 2025.
33. No.
34. SGD 5.35 million.
35. Yes.
36. No.
37. 29 January 2026.
38. No.
39. A misconfigured network policy blocking selected east-west application traffic.
40. No. Revision 4 superseded Revision 3.
41. No.
42. Marcus Lee.
43. Senior Network Engineer.

- 44. No.
- 45. Sarah Wong.
- 46. Sara Wong.
- 47. No.
- 48. Ethan Lim.
- 49. Ethan Lin.
- 50. No.
- 51. 17 minutes.
- 52. Approximately 25%.
- 53. 5.5 minutes.
- 54. 178.
- 55. 31.
- 56. 10.
- 57. Approximately 88%.
- 58. SGD 170,000.
- 59. Approximately 18%.
- 60. Approximately SGD 146,160.
- 61. CW-2025-017.
- 62. One sensor falsely indicated high temperature because it had partially detached, while another sensor and manual measurement showed acceptable cargo temperature.
- 63. Because one faulty sensor can create a false alarm.
- 64. No.
- 65. Authorized Quality Assurance personnel.
- 66. Level 3.
- 67. No.
- 68. AI may provide advice, but disagreement must be investigated using multiple measurements, and authorized humans retain final disposal authority.
- 69. Older documents sometimes had stronger lexical similarity than newer versions.
- 70. Effective-date metadata, status filtering, version-aware retrieval, metadata constraints, reranking, and recency/version logic.
- 71. 86.2%.
- 72. 88.9%.
- 73. 93.7%.
- 74. 94.1%.
- 75. 97.1%.
- 76. 0.79.
- 77. 0.86.
- 78. It increased from 78.6% to 94.2%.
- 79. 300.
- 80. 267.
- 81. No.
- 82. No.
- 83. No.
- 84. Banking changes require independent verification regardless of OCR confidence.
- 85. A fraudulent invoice requested payment to a new bank account; independent verification identified the fraud and no payment was made.

86. Confidence that the value was extracted correctly.
87. No.
88. No.
89. No.
90. No.
91. Query classification -> permission and metadata constraints -> dense and BM25 retrieval -> fusion -> reranking -> evidence selection -> sufficiency checking -> cited answer generation.
92. Yes.
93. Dense retrieval and BM25.
94. Reranking.
95. Evidence sufficiency checking.
96. To reduce unsupported or hallucinated answers.
97. State that sufficient information was not found.
98. No.
99. No.
100. Failure to retrieve evidence means the system lacks support for a conclusion; it does not prove the opposite fact.
101. October 2025.
102. 15 December 2025.
103. 29 January 2026.
104. October was preliminary; 15 December was official production go-live; 29 January completed the remaining Nimbus regional rollout.
105. SGD 4.8 million.
106. SGD 5.35 million.
107. SGD 5.18 million.
108. SGD 812,000.
109. They describe different cost categories and periods: implementation budget versus later operating cost.
110. Nimbus operated a reduced Atlas configuration.
111. No.
112. No.
113. No.
114. No.
115. No.
116. No.
117. No.
118. No.
119. No. It used gradient-boosted decision trees.
120. No.
121. v2.6 is current after 30 September 2025. v2.3 remains relevant to historical pilot performance.
122. D01 contains the original planning figure; the budget was later revised to SGD 5.35 million.
123. Atlas officially entered production on 15 December 2025, while complete regional feature rollout finished later.
124. No.
125. No.
126. Investigate, compare available sensors, inspect refrigeration, take a calibrated manual measurement, document the event, and escalate if uncertainty remains.
127. It demonstrated how a faulty single sensor could produce a false temperature excursion.
128. No.

- 129. Historical investigations or explicit historical queries.
- 130. Historical procedures retain value for incident reconstruction and audit purposes.
- 131. The question is underspecified.
- 132. Different data categories have different retention policies.
- 133. 90 days.
- 134. 400 days.
- 135. Up to seven years where required.
- 136. Approximately 30 days.
- 137. Cargo type and applicable handling requirement.
- 138. Approval depends on workflow, role, risk category, permissions, and business rules.
- 139. Employee role, workflow, risk classification, permissions, and business rules.
- 140. No.
- 141. Inspect refrigeration, compare sensor sources, conduct calibrated manual measurement, record the inspection, and escalate to QA if uncertainty remains.
- 142. No.
- 143. CW-2025-017.
- 144. CC-14 Revision 4.
- 145. Level 3.
- 146. Authorized Quality Assurance personnel.
- 147. No.
- 148. No.
- 149. A calibrated manual temperature measurement.
- 150. Revision 3 permits stronger action from a single sensor, while Revision 4 explicitly accounts for sensor disagreement.
- 151. The provided knowledge base does not contain enough information to answer this question.
- 152. The provided knowledge base does not contain enough information to answer this question.
- 153. The provided knowledge base does not contain enough information to answer this question.
- 154. The provided knowledge base does not contain enough information to answer this question.
- 155. The provided knowledge base does not contain enough information to answer this question.
- 156. The provided knowledge base does not contain enough information to answer this question.
- 157. The provided knowledge base does not contain enough information to answer this question.
- 158. The provided knowledge base does not contain enough information to answer this question.
- 159. The provided knowledge base does not contain enough information to answer this question.
- 160. The provided knowledge base does not contain enough information to answer this question.
- 161. The provided knowledge base does not contain enough information to answer this question.
- 162. The provided knowledge base does not contain enough information to answer this question.
- 163. The provided knowledge base does not contain enough information to answer this question.
- 164. The provided knowledge base does not contain enough information to answer this question.
- 165. The provided knowledge base does not contain enough information to answer this question.
- 166. The provided knowledge base does not contain enough information to answer this question.
- 167. The provided knowledge base does not contain enough information to answer this question.
- 168. The provided knowledge base does not contain enough information to answer this question.
- 169. The provided knowledge base does not contain enough information to answer this question.
- 170. The provided knowledge base does not contain enough information to answer this question.

Optional Evaluation Rubric

Retrieval Accuracy

Measure whether the correct evidence was retrieved. Suggested metrics: Recall@1, Recall@3, Recall@5, Recall@10, and MRR.

Answer Accuracy

Track factual accuracy, numerical accuracy, version-sensitive accuracy, multi-hop accuracy, and unsupported-question accuracy separately.

Hallucination Rate

Use questions 151-170. Hallucination Rate = unsupported questions answered with invented facts / total unsupported questions.

Version Conflict Accuracy

Use questions 21-40, 101-110, and 121-130. The system should prefer newer effective documents unless the query explicitly asks for historical information.

Entity Disambiguation Accuracy

Use questions 41-50 to test whether similar names such as Marcus Lee / Marcus Li, Sarah Wong / Sara Wong, and Ethan Lim / Ethan Lin are kept distinct.

Extreme Configuration

For a harder test, split documents into chunks, remove titles from chunk text, duplicate superseded SOPs, use small top-k retrieval, and disable metadata filtering. This exposes retrieval weaknesses quickly.